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Power Spectral Density of Free-Standing Viscoelastic Films by Adiabatic Approximation

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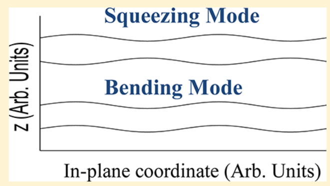
ABSTRACT: In a previous study, we calculated the surface dynamics of noisy viscoelastic supported films by using an adiabatic approximation. An expression was derived for the time-dependent power spectral density (PSD), which was found to produce good agreement with experiment. In this study, we extend the treatment to viscoelastic free-standing films. Two sets of surface capillary normal modes, namely, the squeezing and bending modes, were found. The frequency dispersion relation of the former resembles that of supported films. The latter is distinctively different and diverges at long wavelengths. By incorporating the experimental conditions, we obtained satisfactory agreement between theory and experiment.



INTRODUCTION

As technology strives to make ever smaller devices, materials are molded into ever smaller structures. As the structure sizes are decreased below ~50 nm, interfacial effects start to become important. In the case of polymers, there can be additional confinement effects because the unperturbed size of a polymer chain is typically between 1 and 30 nm. Numerous experiments have shown that the dynamics of polymer films with thickness below 50 nm can be noticeably different from that of the bulk.[1−6] For supported films, namely, those that are supported by a substrate, many studies have found that non-bulk-like dynamics are attributable to interfacial effects.[7−12] For free-standing films, experimental results indicate that other factors may come into play. For example, free-standing polystyrene (PS) films have been found to exhibit two glass-transition temperatures, T_g, both below the bulk value.[13,14] One occurs at a temperature equal to the T_g of the films supported by silicon with half the thickness; the other occurs at a significantly lower temperature that depends on the polymer molecular weight. The former corroborates with the general perception that the dynamics of the films are dominated by the air interface, so free-standing films, which possess twice as many air interfaces as supported films, require less thickness reduction for the same T_g reduction to occur. The latter necessitates a different mechanism that most probably involves the movement of whole chains, which may thereby explain the molecular-weight-dependent T_g. A frequently cited model is the sliding chain model proposed by de Gennes[15] and later elaborated on by Lipson and Milner.[16,17] In this model, the enhanced mobility at an interface could facilitate the enhanced dynamics of the whole film if the chains form large loops at an air interface that extend well into the center of the film or if they make one contact at each of the two air interfaces while traversing the whole film.[15] Recent experiments show that the dynamics (specifically, the viscosity) of polymer films can be effectively revealed by the dynamics of the surface capillary waves.[12,18−22] By fitting the surface dynamics measurement of short-chain, unentangled PS supported by silicon with different thicknesses to the surface capillary wave theory of noisy Newtonian liquid films, Yang et al. deduced that the films could be treated as a bulklike layer residing beneath a mobile layer that exhibited Arrhenius dynamics.[12,23] More recently, the study was extended to entangled, viscoelastic PS-supported films,[21,24] where the surface dynamics data were found to fit well to an expression derived for the time-dependent power spectral density (PSD) of noisy viscoelastic films by using an adiabatic approximation.[25] In this study, we investigate the surface dynamics of viscoelastic free-standing films by applying the same adiabatic approximation to the surface capillary normal modes of free-standing liquid films, derived from continuum hydrodynamic theory.[26] Previously, Vrij[27] derived the squeezing surface capillary modes, where the surface undulations at the top and bottom interfaces are 180° out of phase (Figure 1) and have a dispersion relation similar to that of supported films. The similarity originates from the fact that the fluid flow pattern within the upper or lower half of the films in these modes is identical to that of a supported film with half the film thickness. In this study, we find there to be additional bending modes, where the surface undulations at the top and bottom interfaces are in phase (Figure 1). By the adoption of the experimental conditions, an expression for the time-dependent PSD of the top surface is derived and is found to produce satisfactory agreement with experiment.

SURFACE DYNAMICS OF PURELY VISCOUS FREE-STANDING LIQUID FILMS

Normal Modes of the System. Consider a free-standing film with thickness $2d$ as shown in Figure 1. We use $h_1(\vec{r}, t)$ and

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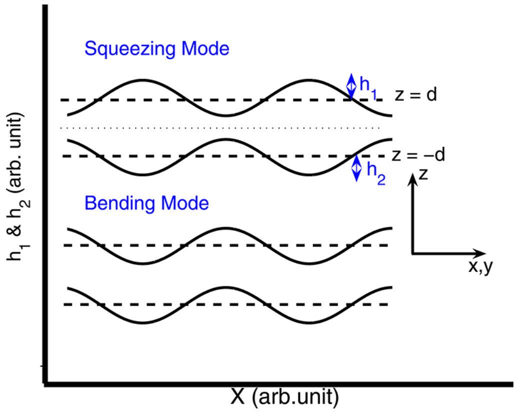

Figure 1. Illustration of the phase relation between the top (h_1) and bottom (h_2) surface undulations in a free-standing film with squeezing (top) and bending (bottom) surface capillary wave excitation.

$h_2(\vec{r}, t)$ to denote the undulation of the top and bottom surfaces about the average positions $z = d$ and $z = -d$, respectively. Here, $\vec{r} = (x, y)$ denotes a vector in the plane of the film and t is time. We have chosen the surface normal to lie in the z direction and the film midplane to coincide with the $z = 0$ plane. Because of the mirror symmetry about the midplane, this choice renders simple relations $h_1(\vec{r}, t) = h_2(\vec{r}, t)$ and $h_1(\vec{r}, t) = -h_2(\vec{r}, t)$ for the bending and squeezing modes, respectively (Figure 1). We denote $h^{\mathrm{b}}(\vec{r}, t)$ and $h^{\mathrm{s}}(\vec{r}, t)$ as the undulation of the bending and squeezing modes, respectively, so that $h_1(\vec{r}, t) = h^{\mathrm{b}}(\vec{r}, t) + h^{\mathrm{s}}(\vec{r}, t)$ and $h_2(\vec{r}, t) = h^{\mathrm{b}}(\vec{r}, t) - h^{\mathrm{s}}(\vec{r}, t)$. Working in Fourier space, we write

$$\begin{aligned} h_{1,2}(\vec{r}, t) &= \frac{1}{\sqrt{A}} \sum_{\vec{q}} h_{1,2}(\vec{q}, t)\, \mathrm{e}^{\mathrm{i}\vec{q}\cdot\vec{r}} \\ h^{\mathrm{b,s}}(\vec{r}, t) &= \frac{1}{\sqrt{A}} \sum_{\vec{q}} h^{\mathrm{b,s}}(\vec{q}, t)\, \mathrm{e}^{\mathrm{i}\vec{q}\cdot\vec{r}} \end{aligned} \tag{1}$$

where A denotes the surface area of the film and $\vec{q} = (q_x, q_y)$ is the in-plane wave vector. We shall write $q = |\vec{q}|$ below.

Equations of Motion. To establish the equations of motion for $h_{1,2}(\vec{r}, t)$ and $h^{\mathrm{b,s}}(\vec{r}, t)$, we utilize the following relation

$$\partial_t h_{1,2}(\vec{r}, t) = v_z(\vec{r}, z, t)|_{z=\pm d} \tag{2}$$

where $v_z(\vec{r}, z, t)$ is the z component of the velocity field $\vec{v}(\vec{r}, z, t) = (v_x(\vec{r}, z, t), v_y(\vec{r}, z, t), v_z(\vec{r}, z, t))$ and the geometry shown in Figure 1 is assumed.

We apply the Navier–Stokes equation (NSE) to the film, which is presumed to be an incompressible fluid, to find $\vec{v}(\vec{r}, z, t)$. Pertinent to the experimental conditions, we shall neglect the nonlinear and inertial terms as in ref 26. Without a loss of generality (Appendix A), hereafter we confine the discussion to the case where $v_y = 0$, $q_y = 0$, and $q_x = q$. On the basis of previous work, the general solutions to the NSE are given by[26]

$$\begin{aligned} v_z = {} & (C_3 + C_2 qz)\cosh(qz) + (C_4 - C_2 + C_1 qz) \\ & \sinh(qz),\ v_x = \frac{\mathrm{i}}{q}\partial_z v_z \end{aligned} \tag{3}$$

where $C_{1,2,3,4}$ represents coefficients to be determined by the boundary conditions (BC). With the aid of the pressure field, $P = \eta(q^2\partial_z^{\ 3} v_z - \partial_z v_z)$, and the internal stress tensor, $\sigma_{ij} = \eta(\partial_i v_j + \partial_j v_i) - P\delta_{ij}$, where η denotes the shear viscosity, the BC can be written as

$$\sigma_{xz}|_{z=\pm d} = 0,\ \sigma_{zz}|_{z=\pm d} \mp \frac{\delta F[h_{1,2}(q)]}{\delta h_{1,2}(q)} = 0 \tag{4}$$

where $F[h_{1,2}(q)]$ is the change in the free energy of the film arising from the surface undulations. Up to quadratic orders of the undulation amplitudes, the former is given by $F[h_{1,2}(q)] = (^1/_2)\sum_q \Phi_q(|h_1(q)|^2 + |h_2(q)|^2) = \sum_q \Phi_q(|h^{\mathrm{b}}(q)|^2 + |h^{\mathrm{s}}(q)|^2)$. In eq 4, the first relation warrants the vanishing of the internal shear stress at either film interface; the second one expresses the normal stress balance. We consider the film thickness regime where the surface tension dominates the dispersion energy so that $\Phi_q \approx \gamma q^2$. By combining eqs 2–4, one arrives at

$$\partial_t h^\iota(q, t) = -\Gamma_q^\iota h^\iota(q, t) + \xi_q^\iota(t),\ \iota = \mathrm{b, s} \tag{5}$$

where $\Gamma_q^\iota = [(\Phi_q/2\eta)(1/\sum_\iota(q))]$. As will be apparent below, it is the relaxation rate of the surface capillary normal mode, ι. The second term is the thermal noise whose properties are to be specified later. Besides, $\sum_{\mathrm{b}}(q) = ((q^2 d(s^2 - c^2) + qcs)/c^2)$ and $\sum_{\mathrm{s}}(q) = ((q^2 d(c^2 - s^2) + qcs)/s^2)$, with $c = \cosh(qd)$ and $s = \sinh(qd)$. In the long-wavelength limit, $qd < 1$, which is valid in typical experiments, and the relaxation rates of the bending and squeezing modes are given by

$$\Gamma_q^{\mathrm{b}} \approx \frac{\Phi_q d}{4\eta}\frac{3}{(qd)^4} \quad \text{and} \quad \Gamma_q^{\mathrm{s}} \approx \frac{\Phi_q d}{4\eta} \tag{6}$$

Note that the bending mode relaxes much faster than the squeezing mode in this limit and in fact Γ_q^{b} diverges as $q \to 0$. Note also that Γ_q^{s} is the same as the relaxation rate of a supported film with thickness d in the presence of a significant slip length.[26]

■ VISCOELASTIC EFFECTS BY ADIABATIC APPROXIMATION

In the last section, we derived that there are two sets of surface capillary normal modes associated with a purely viscous free-standing film and that each has a distinctly different dispersion relation. In this section, we shall investigate the influences of viscoelasticity on these modes. Viscoelaticity gives rise to extra modes that are primarily associated with motions of the elastic component.[26] Usually, these extra modes have much faster dynamics than the purely viscous ones. This separation of time scales validates treatments based on the adiabatic approximation, as previously introduced by Lam et al.[25] Following their approach,[28] we note that the surface undulation of the films, $h^\iota(q)$, with $\iota = \mathrm{b, s}$, consists of two components, namely, a viscous one $H^\iota(q)$ and an elastic one $u^\iota(q)$. Together, they give $h^\iota(q,t) = H^\iota(q,t) + u^\iota(q,t)$. The free energy is then given by

$$F \equiv F_{\mathrm{b}} + F_{\mathrm{s}},\ F_\iota = \sum_q [\Phi_q|h^\iota(q)|^2 + \Psi_q^\iota|u^\iota(q)|^2] \tag{7}$$

Because of the mirror symmetry, couplings between the bending and squeezing modes are prohibited up to quadratic order. The prefactors, Ψ_q^ι, are related to the high-frequency elastic modulus of the film originating from interactions at atomic separations. For thin elastic plates,[29]

$$\Psi_q^{\mathrm{b}} = \kappa q^4 \quad \text{where} \quad \kappa = \frac{Yd^3}{3(1-\nu^2)} \tag{8}$$

Here, Y is the high-frequency Young modulus and ν is the Poisson ratio. To find Ψ_q^s, we note that a film in a squeezing mode excitation has its midplane still and so within either the upper or the lower half of the film the physics is identical to that of a supported film under a significant slip length. Accordingly, Ψ_q^s can be worked out by calculating the energy required to deform a supported film with thickness d. Such a calculation has been performed by Fredrickson et al.,[30] who found

$$\Psi_q^s = \varsigma q^{-2} \quad \text{where} \quad \varsigma = \frac{3\mu_0}{4d^3} \tag{9}$$

In this equation, μ_0 is the high-frequency shear modulus and is related to the viscoelastic time, defined here to be $\tau_{ve} = (\eta/\mu_0)$. In polymers, τ_{ve} is the reptation time upon which an entangled polymer transitions from the elastic rubbery state to the viscous liquid state.

In the present adiabatic approximation, the elastic component is presumed to be swift enough to be able to equilibrate itself to elastic configurations around the minimum free energy state given any instantaneous value of the viscous component. Formally, one writes $u^\iota(q) = u_0^\iota(q) + \delta u^\iota(q)$, where $u_0^\iota(q)$ takes on values that minimize F. One can show that energy minimization leads to

$$u_0^\iota(q) = -\frac{\Phi_q}{\Phi_q + \Psi_q^\iota} H^\iota(q) \tag{10}$$

The second part of $u^\iota(q)$, namely, $\delta u^\iota(q)$, represents thermally activated elastic vibrations about $u_0^\iota(q)$. In general, it bears the time dependence of a damped oscillator under thermal noise. We approximate their relaxation time τ_ι^e by the period of the free oscillator. From elasticity theory, $\tau_\iota^e = ((\rho_A)/(\Psi_q^\iota))^{1/2}$,[29] where $\rho_A = 2d{\cdot}\rho$ denotes the area density of the film.

From the foregoing discussions, one may write

$$h^\iota(q) = H^\iota(q) + u_0^\iota(q) + \delta u^\iota(q) = \frac{\bar{\Phi}_q^\iota}{\Phi_q} H^\iota(q) + \delta u^\iota(q) \tag{11}$$

where use has been made of eq 10 in obtaining the second equality and $\bar{\Phi}_q^\iota = ((\Phi_q)\Psi_q^\iota)/(\Phi_q + \Psi_q^\iota)$. Substituting eqs 10 and 11 into the free-energy expression in eq 7, we find

$$F_\iota \approx \sum_q [\bar{\Phi}_q^\iota |H^\iota(q)|^2 + (\Phi_q + \Psi_q^\iota)|\delta u^\iota(q)|^2] \tag{12}$$

where H^ι and δu^ι are now decoupled. It is clear that the viscous components, $H^\iota(q, t)$ obey an equation of motion similar to eq 5:

$$\partial_t H^\iota(q, t) = -\bar{\Gamma}_q^\iota H^\iota(q, t) + \xi_q^\iota(t) \quad \text{where} \quad \bar{\Gamma}_q^\iota = \Gamma_q^\iota \frac{\bar{\Phi}_q^\iota}{\Phi_q} \tag{13}$$

From this equation, it is apparent that the impacts of viscoelasticity are contained in the screening factors, $\alpha_\iota \equiv ((\bar{\Phi}_q^\iota)/\Phi_q) \leq 1$. A plot of α_ι versus q using realistic material parameters appropriate to the film used in this study (see below) is shown in Figure 2. When $\alpha_\iota < 1$, surface undulations due to viscous flow are partially smoothed out instantaneously (i.e., screened) by elastic deformations and hence decay at a reduced rate $\bar{\Gamma}_q^\iota = \alpha_\iota \Gamma_q^\iota$. For the squeezing modes, eq 9 shows that they are elastically rigid at small q. We get $\alpha_s = 1/(1 + q^4/q_s^4)$, where $q_s = (\varsigma\gamma^{-1})^{1/4}$ denotes a crossover point. We estimate that $q_s \approx 100\ \mu m^{-1}$, which is beyond the range of q typically accessed in an experiment.[12,18−20] We thus have $q \ll q_s$ and $\alpha_s \approx 1$, indicating weak screening. Therefore, the experimentally probed squeezing mode is expected to behave like those found in purely viscous films. However, the bending modes are elastically much softer at small q. Specifically, $\alpha_b = 1/(1 + q_b^2/q^2)$, with $q_b = (\gamma\kappa^{-1})^{1/2}$. We estimate $q_b \approx 3\ \mu m^{-1}$, which lies well within the range of q accessed in the experiment. Therefore, for a range of values of $q < q_b$ of experimental relevance, α_b is small and tends to zero in the small q limit, implying strong screening.

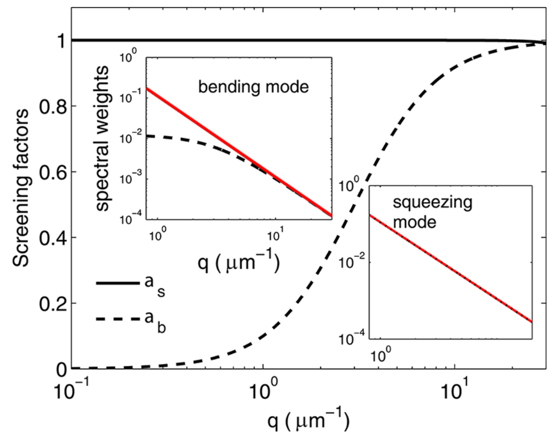

Figure 2. (Main panel) The screening factors α_ι plotted against q. (Left inset) Spectral weights vs q of the unscreened (solid line, $(k_BT)/2\Phi_q$) and screened (dashed line, $\alpha_\iota(k_BT/2\Phi_q)$) bending modes. (Right inset) Corresponding spectral weight plots of the squeezing mode. A strong suppression of the screened bending mode is apparent in the small q region.

THEORETICAL PSD EVOLUTION AND COMPARISON WITH EXPERIMENT

Theoretical PSD Evolution. Here, we focus on the evolution of the PSD of the top surface, $A_q(t) = \langle|h_1(q, t)|^2\rangle$ ($\langle...\rangle$ denotes an average over the thermal noise configurations), which is the one experimentally accessible by atomic force microscopy (AFM). Using $h_1(q) = h^b(q) + h^s(q)$ and neglecting correlations between the normal modes (Appendix B), one can write

$$A_q(t) = A_q^s(t) + A_q^b(t) \tag{14}$$

where $A_q^\iota(t) = \langle|h^\iota(q, t)|^2\rangle$. By using eq 11, one can write

$$A_q^\iota(t) = \left(\frac{\bar{\Phi}_q^\iota}{\Phi_q}\right)^2 \langle|H^\iota(q, t)|^2\rangle + \langle|\delta u^\iota(q, t)|^2\rangle \tag{15}$$

where the cross term between H^ι and δu^ι is absent because they are not coupled in the free-energy expression (eq 12 and ref 25). Pertinent to the presumption of the adiabatic approximation, relaxations of the elastic vibrations δu^ι occur on a time scales of τ_ι^e, which is much shorter than $\tau_\iota = 1/\bar{\Gamma}_q^\iota$. Therefore, for any instantaneous value of H^ι, δu^ι can be regarded as being at thermodynamic equilibrium. By using eq 12 and the equipartition law, we then arrive at

$$\langle|\delta u^\iota(q, t)|^2\rangle = \frac{k_BT}{2(\Phi_q + \Psi_q^\iota)} \tag{16}$$

Following ref 25, the equipartition law also implies that the noise $\bar{\xi}_q^\iota(t)$ defined in eq 13 has the following property,

$$\langle \bar{\xi}_q^{\iota *}(t)\bar{\xi}_{q'}^{\iota}(t')\rangle = 2D_q^\iota \delta_{q,q'}\delta(t-t') \tag{17}$$

where ι = b, s and $D_q^\iota = \bar{\Gamma}_q^\iota k_B T[2\bar{\Phi}_q^\iota]^{-1}$.

By applying this relation to eq 13 as in ref 25, we find

$$\langle |H^\iota(q,t)|^2\rangle = \langle |H^\iota(q,t=0)|^2\rangle e^{-2\bar{\Gamma}_q^\iota t} + \frac{k_B T}{2\bar{\Phi}_q^\iota}(1 - e^{-2\bar{\Gamma}_q^\iota t}) \tag{18}$$

When eqs 14–18 are combined, a full expression of $A_q(t)$ can be established. One may perceive that it gives $A_q(t = \infty) = k_B T/\Phi_q$, which is consistent with the expectation from the free-energy expression in eq 7. However, this ideal result may not be used to compare with experiment if the temporal resolution τ_{res} of the experimental probe is limited. Specifically, the time scales of the elastic vibrations, τ_ι^e, are usually much shorter than τ_{res}. (See below.) Under this circumstance, the experimental probe would detect the average value of δu^ι (i.e., $\overline{\delta u^\iota}(q, t)$), which vanishes for $\tau_{res} > \tau_\iota^e$. Consequently, the PSD, as seen by the probe, will amount to only the first term in eq 15, and the apparent experimental PSD, $A_q'(t)$, would be

$$A'_q(t) = \left(\frac{\bar{\Phi}_q^b}{\Phi_q}\right)^2 \langle |H^b(q,t)|^2\rangle + \left(\frac{\bar{\Phi}_q^s}{\Phi_q}\right)^2 \langle |H^s(q,t)|^2\rangle \tag{19}$$

On inserting eq 18, we get

$$A'_q(t) = A_q^s(0)e^{-2\bar{\Gamma}_q^s t} + \alpha_s \frac{k_B T}{2\Phi_q}(1 - e^{-2\bar{\Gamma}_q^s t}) + A_q^b(0)e^{-2\bar{\Gamma}_q^b t} + \alpha_b \frac{k_B T}{2\Phi_q}(1 - e^{-2\bar{\Gamma}_q^b t}) \tag{20}$$

As shown above, $\alpha_s \approx 1$, leading to $\bar{\Gamma}_q^s \approx \Gamma_q^s$. However, α_b steadily approaches zero as q decreases below q_b, leading to pronounced screening and hence a strong suppression of the contribution from the bending mode to the PSD (Figure 2). To visualize this effect, we plot the predicted time sequence of $A_q(t)$ and $A_q'(t)$ assuming the experimental parameters used in this work (see the next section) and $A_q^\iota(0) = 0$ in Figure 3a,b, respectively. Evidently, $A_q(t)$ and $A_q'(t)$ begin to depart from one another below $q_b \approx 3$ μm $^{-1}$. In the next section, we shall see that $A_q'(t)$, not $A_q(t)$, is what describes the experimental observation confirming the alleged effect of the temporal resolution of the measurement probe.

Comparison with Experiment. In this section, we shall demonstrate how we apply the above result to the experimental PSD shown in Figure 4 (open circles). These data had been taken from a moderately entangled poly(methyl methacrylate) (PMMA) free-standing film with a molecular weight, MW, of 56 kg/mol (the entanglement MW of PMMA is about 10 kg/mol[31]) and a film thickness of 73 nm (d = 36.5 nm). To prepare the free-standing films, a water transfer technique was used.[32] In brief, a solution of PMMA in toluene was spun cast on mica. The resultant film was floated on deionized water and subsequently scooped with a transmission electron microscopy grid. During the measurement, the film was annealed at T = 443 K. At this temperature, the film was unstable and roughened with time. This permits us to compare the surface dynamics of the film with the theory. At different cumulative times, t, the film was quenched within a short period (denoted by $\tau_{quench} \approx 1$ s) to room temperature, whereupon its PSD was measured by tapping-mode AFM. Further details of the measurement can be found in refs 21–24. As pointed out above, AFM measures the top surface of the film. Therefore, only the PSD of the top surface, namely, $A_q(t) = \langle |h_1(q,t)|^2\rangle$, is detected.

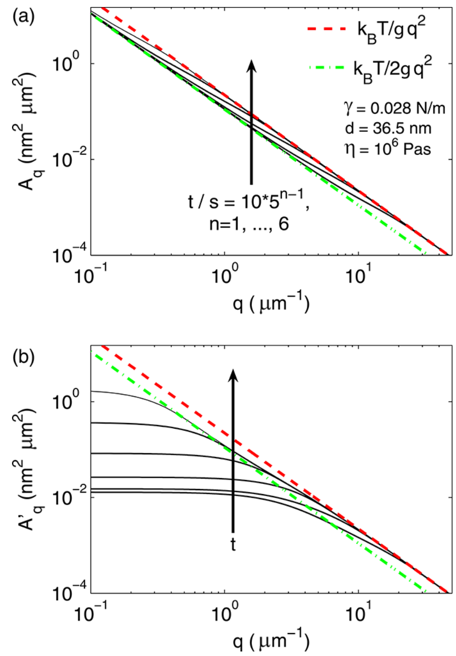

Figure 3. (a) Plot of the calculated sequence of the ideal PSD $A_q(t)$ obtained by assuming parameters appropriate to the films discussed in the following section and $A_q^\iota(0) = 0$. (b) Corresponding plot of the apparent PSD $A_q'(t)$.

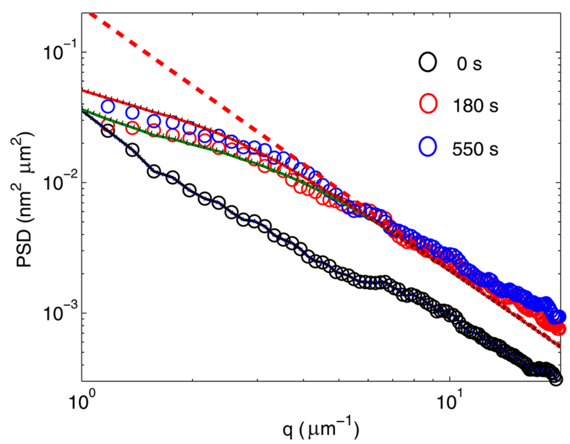

Figure 4. Comparison between theory (eq 20, dotted lines; eq 21, solid lines) and experiment (circles). The parameters and initial conditions are given in the text. The dashed line represents the equilibrium PSD of a liquid film with surface tension γ = 0.028 N/m.

Before going on, we estimate the relevant time scales. First, we consider the relaxation times of the bending and squeezing modes for liquid films, which are $(1/\Gamma_q^b)$ and $(1/\Gamma_q^s)$, respectively (eq 6). By substituting $\eta = 10^6$ Pa s (at T = 443 K) and γ = 0.028 N/m, we find $(1/\Gamma_q^b)$ = 2.3 ms/(q^2 μm^2) and $(1/\Gamma_q^s)$ = 3900 s/(q^2 μm^2). Next, we incorporate the effect of viscoelasticity. As eq 13 shows, viscoelasticity modifies $(1/\Gamma_q^\iota)$ to $\tau_\iota = 1/\bar{\Gamma}_q^\iota = ((\alpha_\iota^{-1})/(\Gamma_\iota^q))$ with ι = b, s. To estimate α_ι, we first calculate q_ι. By assumming that the high-frequency Young modulus $Y = 0.1$ GPa, shear modulus $\mu_0 = 1$ GPa, and Poisson ratio $\nu = 0.3$, we find $\kappa = 1.78$ nN μm and $\varsigma = 1.5 \times 10^4$ Pa/nm^3. These lead to $q_s \approx 100\ \mu\text{m}^{-1}$ and $q_b \approx 3\ \mu\text{m}^{-1}$. Because $q < \sim 30\ \mu\text{m}^{-1}$ (Figure 3), $\alpha_s^{-1} = 1 + (q^4/q_s^4) \approx 1$ and $\alpha_b^{-1} = 1 + (q_s^2/q^2)$. Correspondingly, $\tau_s \approx (1/\Gamma_q^s) = 3900\ \text{s}/(q^2\ \mu\text{m}^2)$ and $\tau_b = 2.3\ \text{ms}/(q^2 + q_b^2)\ \mu\text{m}^2$. The latter shows that the relaxation time of the bending mode, τ_b, lengthens with the introduction of viscoelasticity. Nevertheless, it is still rather short, only ∼23 ms at $q = 1\ \mu\text{m}^{-1}$, where the evolution of the PSD is prominent. However, $\tau_b \approx 3900$ s at $q = 1\ \mu\text{m}^{-1}$. Below we shall see that these vastly different times have a nontrivial impact on the observed PSD. Last, we estimate the time scales of the elastic vibrations, τ_i^e. As noted above, $\tau_i^e = (\rho_A/\Psi_q^i)^{1/2}$. By using $\rho = 1$ g cm^{-3} and $\rho_A = 2d\cdot\rho$, we find that $\tau_b^e \approx 64\ \text{ns}/(q\ \mu\text{m})$ and $\tau_s^e \approx 2.2$ ps q μm. In Figure 5, we plot all these time scales and the viscoelastic time $\tau_{ve} = (\eta/\mu_0) = 1$ ms for the range of q studied in the experiment. As seen, both τ_e^b and τ_s^e are significantly shorter than their viscous counterparts, τ_b and τ_s. This justifies the present adiabatic approximation. In addition, the resonance frequency of the AFM cantilever was 21 kHz ([∼5 × 10^{-5} s]$^{-1}$), making it a slow probe in view of the rapid elastic vibrations. With these considerations only, eq 20 would be adequate for understanding the experimental result shown in Figure 4.

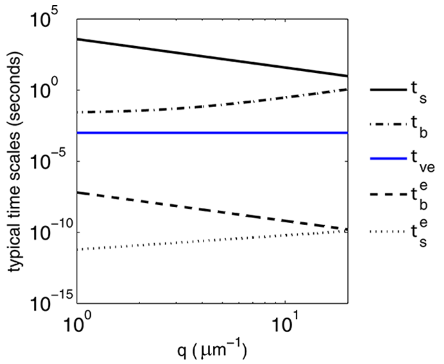

Figure 5. Various time scales involved in the theory and experiment. All are evaluated at 443 K.

We also notice that the use of quenching in the experiment introduces yet another time scale, $\tau_{quench} \approx 1$ s. To complete the discussion, we show that quenching the sample before each AFM measurement leads to a negligible impact. We observe that the bending mode of the viscous vibrations has a relaxation time, τ_b, that can be much shorter than or comparable to τ_{quench} (Figure 5). When $\tau_b < \tau_{quench}$, the bending modes can be taken to be thermally equilibrated at the instantaneous temperature at which quenching took place. Note that τ_b is proportional to the shear viscosity η, which strongly increases as the temperature drops toward the glass-transition temperature of the specimen (∼390 K). Therefore, one expects there to be a temperature, T_{eff} ($\le T$), at which $\tau_b|_{T=T_{eff}} = \tau_{quench}$. This would be the temperature of the bending modes contributing to the PSD as seen in experiments. We revise eq 20 as follows to incorporate this effect

$$A_q'(t) = A_q^s(0)e^{-2\Gamma_q^s t} + \frac{k_B T}{2\Phi_q}(1 - e^{-2\Gamma_q^s t}) + A_q^b(0)e^{-2\bar{\Gamma}_q^b t} + \frac{1}{1 + \frac{q_b^2}{q^2}} \frac{k_B T_{eff}}{2\Phi_q}(1 - e^{-2\bar{\Gamma}_q^b t}) \qquad (21)$$

where $\alpha_s \approx 1$ has been used. By using the measured η versus T of the film, we estimate the values of T_{eff} for various possible values of $\tau_{quench} = 1$ s, 10 s, and 1 min, as shown in Figure 6. We find that all three choices give quantitatively similar results. In the following discussion, we use $\tau_{quench} = 1$ s in making a comparison with experiments.

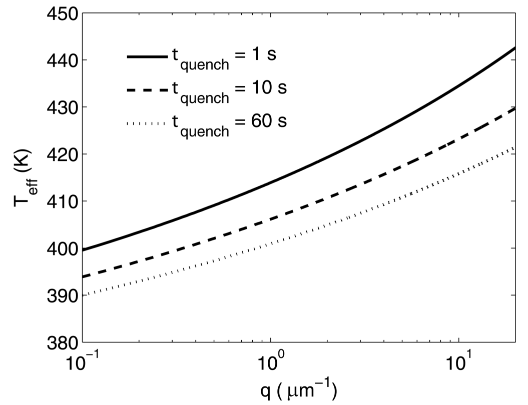

Figure 6. Effective temperature, T_{eff}, at which the bending mode gets frozen in during the quenching process.

Next, we compare the above results with experiment. To demonstrate that quenching makes little impact on the PSD evolution, we shall use both eqs 20 and 21 in calculating the theoretical predictions. To apply these equations, we assume that $h_2(q, t = 0) \approx 0$ because in the fabrication process the film was first spin-coated onto a smooth substrate surface before it was lifted and suspended in a free-standing film. This procedure leaves the bottom surface of the film relatively smooth. With this approximation, we have $A_q^s(0) = A_q^b(0) = A_q(0)/2$. The result obtained by substituting these initial values into eqs 20 and 21 is displayed in Figure 4 by dotted and solid lines, respectively. As seen, the two predictions almost overlap and agree well with experiment. The similarity between the two predictions can be understood by examining the plots of T_{eff} shown in Figure 6. There, one sees that T_{eff} deviates by <13% from T (= 443 K). As shown by eqs 20 and 21, quenching should have an insignificant effect on the measured PSD as observed.

CONCLUSIONS

We have found there to be two sets of normal modes that contribute to the PSD of free-standing liquid films, namely, the bending modes and squeezing modes. At long wavelengths, their relaxation times differ considerably. In the purely viscous case, the modes contribute equally to the PSD. The squeezing mode evolves much more slowly than the bending modes and can lead to the experimentally observable evolution of the PSD. We have also investigated viscoelastic liquid films. Two additional sets of purely elastic bending and squeezing modes arise. Their much shorter relaxation times decouple them from the viscous modes. The elastic squeezing modes are rigid and contribute negligibly to the overall dynamics. By contrast, the much softer elastic bending modes substantially slow the evolution of the viscous counterparts by introducing screening effects. This alters the PSD significantly. They nevertheless do not contribute directly to the apparent PSD measured by the slow experimental probe. We have applied the theory to a slightly entangled free-standing PMMA film, and satisfactory agreement is found.

APPENDIX A

Here we show that only the longitudinal velocity is coupled to v_z. By neglecting the nonlinear and inertial terms, the NSE is simplified to $\eta\nabla^2\vec{v}(\vec{r}, z, t) = \vec{\nabla}P(\vec{r}, z, t)$. In the $\vec{q}$ representation (eq 1), it can be rewritten as

$$\mathrm{i}\vec{q}P(\vec{q}, z) = \eta(\partial_z^{\,2} - q^2)\vec{v}_{\|}(\vec{q}, z),\ \partial_z P(\vec{q}, z) = \eta(\partial_z^{\,2} - q^2)\, v_z(\vec{q}, z) \tag{22}$$

where $\vec{v}_{\|}(\vec{q}, z) \equiv (v_x(\vec{q}, z)$ and $v_y(\vec{q}, z))$. Next, we resolve $\vec{v}_{\|}(\vec{q}, z)$ into longitudinal and transverse components, namely, $\vec{v}_{\|}(\vec{q}, z) = v_{\mathrm{L}}(\vec{q}, z)\hat{\mathrm{e}}_{\mathrm{L}} + v_{\mathrm{T}}(\vec{q}, z)\hat{\mathrm{e}}_{\mathrm{T}}$, where the longitudinal unit vector is defined as $\hat{\mathrm{e}}_{\mathrm{L}} = \vec{q}/q$ and $\hat{\mathrm{e}}_{\mathrm{T}}\cdot\hat{\mathrm{e}}_{\mathrm{L}} = 0$. In addition, $v_{\mathrm{L,T}}(\vec{q}, z) = \hat{\mathrm{e}}_{\mathrm{L,T}}\cdot v_{\|}(\vec{q}, z)$. With these, the first term in eq 22 can be rewritten as

$$\mathrm{i}qP(\vec{q}, z) = \eta(\partial_z^{\,2} - q^2)\, v_{\mathrm{L}}(\vec{q}, z),\ 0 = \eta(\partial_z^{\,2} - q^2)\, v_{\mathrm{T}}(\vec{q}, z) \tag{23}$$

These expressions show that v_{L} and v_z are coupled to each other via the pressure field but are uncoupled to v_{T}. Moreover, only the magnitude of q enters the equations as expected from the in-plane rotational symmetry of the problem.

APPENDIX B

Here we show that the cross term between the normal modes can be neglected, as we did in eq 14. It can be written as $C_q(t) = 2Re\{\langle h^{\mathrm{b}*}(q, t)\, h^{\mathrm{s}}(q, t)\rangle\}$. At equilibrium, it must vanish because of the symmetries of the b and s modes, namely, $C_q(t = \infty) = 0$. From eq 11, we write $h^\iota(q, t) = \alpha_\iota H^\iota(q, t) + \delta u^\iota(q, t)$. Because $\delta u^\iota(q, t)$ can be regarded as being in instaneous thermodynamic equilibrium, $\langle \delta u^{\mathrm{b}}*(q,\ t)\ \ \delta u^{\mathrm{s}}(q,\ t)\rangle = 0$. Thus, $C_q(t) \propto Re\{\langle H^{\mathrm{b}}*(q, t)\ H^{\mathrm{s}}(q, t)\rangle\}$. From eq 13, we have

$$H^\iota\left(q, t\right) = H^\iota\left(q, t = 0\right)\mathrm{e}^{-\overline{\Gamma}^\iota_q t} + \int_0^t \mathrm{e}^{-\overline{\Gamma}^\iota_q(t-t')}\overline{\xi}^\iota_q\left(t'\right)\mathrm{d}t'$$

Making use of the noise properties in eq 17, we then find $C_q(t) \approx \mathrm{e}^{-(\overline{\Gamma}^{\mathrm{b}}_q+\overline{\Gamma}^{\mathrm{s}}_q)t}$. Because $1/\overline{\Gamma}^{\mathrm{b}}_q \approx 1$ ms, which is much shorter than typical experimental times, the cross term can then be dropped.

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Notes

The authors declare no competing financial interest.

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